

Mind-Controlled PROSTHETIC ARM

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At times, during multitasking, you would have wished to have more than two hands attached to your body, which you could control using your mind—just like you control your natural hands. But considering it as unrealistic, and only belonging to sci-fi movies, you would have immediately dismissed the idea.

What if I tell you that it can actually be done? With the help of brain

waves, you can now control any prosthetic or robotic hand. Imagine operating several of such artificial body parts with your thoughts. Isn't it amazing?

By adding pneumatic actuators, you can even provide extra power to lift heavy loads without pushing any button. All you need is brainpower. Thanks to AI the possibilities do not end here. You can apply your brainpower various other ways and

explore further.

All this sounds so cool! You would be eager to know more. So, without further ado, let us begin the awesome journey of building this project by collecting the following material.

REQUIRED MATERIAL

Component name	Quantity	Description
Raspberry Pi 4	1	2GB or greater RAM
Prosthetic hand	1	3D printed prosthetics
Servo motor	4	12kg-cm torque
EEG headset	1	Mindwave mobile 2

Optional: Voice bonnet to direct servo connection and control.

Assembly of prosthetic hand

There are various open source prosthetic body parts available in the market. If you do not want to purchase them, you can 3D print them at home. I used a robotic arm made by InMoov. The complete prosthetic/robotic hand after assembling the required parts, including servo motors and fingers, looks as shown in Fig. 1.



Fig. 1: Author's prototype



Fig. 2: Robot hand base part



Fig. 3: Attaching servo motor to move fingers